

PCB point-to-point resistance report

Board RefPowerRankin jointCPS385v08.kicadpcb

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Assumptions

Setting	Value
Barrel plating	18 um (0.517 oz-equivalent)
Hole value means	bit
Barrel length	centre
Outer-layer plating adds	True
Zone model	ladder
Ambient	25 C
Default current	1 A
Default signal voltage	3.3 V

DC only. No skin effect, no self-heating, no etch taper. Resistance is reported at 20 C and at ambient. Per-pair voltage or current overrides are marked in the summary.

Stackup

Layer	Finished	oz	z centre
F.Cu	53 um	1.523	0.0085 mm
In1.Cu	35 um	1.006	0.1525 mm
In2.Cu	35 um	1.006	1.4275 mm
B.Cu	53 um	1.523	1.5715 mm

Net selection

Reporting on 1 net(s): /Reference Design/PACK_P

Explicitly ignored: GNDREF

Summary

Net	From	To	Zone	R @20 C	R @ambient	V in	I	Drop	V out	Parallel gain
/Reference Design/PACK_P	J11.B	U9.D	ladder	5.287 mohm	5.391 mohm	3.3 V	1 A	5.391 mV	3.2946 V	23.3%
/Reference Design/PACK_P	J11.B	C17.1	ladder	5.285 mohm	5.389 mohm	3.3 V	1 A	5.389 mV	3.2946 V	23.3%
/Reference Design/PACK_P	J11.B	C16.1	ladder	5.255 mohm	5.359 mohm	3.3 V	1 A	5.359 mV	3.2946 V	23.2%
/Reference Design/PACK_P	J11.B	R8.1	ladder	6.391 mohm	6.516 mohm	3.3 V	1 A	6.516 mV	3.2903 V	20.3%
/Reference Design/PACK_P	J11.B	C44.1	ladder	5.129 mohm	5.23 mohm	3.3 V	1 A	5.230 mV	3.2948 V	23.6%
/Reference Design/PACK_P	J11.B	CP2.1	ladder	5.184 mohm	5.286 mohm	3.3 V	1 A	5.286 mV	3.2947 V	23.4%
/Reference Design/PACK_P	J11.B	U2.CSP2	mesh 0.25mm	195.5 mohm	199.4 mohm	3.3 V*	0.001 A*	0.199 mV	3.2998 V	14.9%
/Reference Design/PACK_P	J11.B	U2.BAT	mesh 0.25mm	96.9 mohm	98.8 mohm	3.3 V*	0.001 A*	0.099 mV	3.2999 V	25.5%
/Reference Design/PACK_P	J11.B	U11.D	ladder	5.36 mohm	5.465 mohm	3.3 V	1 A	5.465 mV	3.2945 V	21.5%
/Reference Design/PACK_P	J11.B	C15.1	ladder	189.5 mohm	193.2 mohm	3.3 V*	0.05 A*	9.661 mV	3.2903 V	0.9%
/Reference Design/PACK_P	J11.B	C39.2	ladder	90.57 mohm	92.35 mohm	3.3 V*	0.05 A*	4.618 mV	3.2954 V	1.8%

* marks a value overridden for that pair rather than taken from the operating-condition defaults.

Solved in 0.48 s total (scipy backend).

Detail

/Reference Design/PACK_P (net 10, 12 terminals)

- Via array: 13 vias, centroid (117.3462, 94.8077), extent 5.630 mm
- Via array: 7 vias, centroid (117.3, 99.1), extent 4.200 mm
- pour aspect ratio is only 1.9:1 - a 1-D strip model is questionable on copper this square; consider the mesh model
- pad C17.1 on F.Cu has no track and sits 0.150 mm outside the filled copper (zone clearance / thermal relief); connected to the pour. The relief spokes themselves are not modelled, so this pad reads slightly optimistic
- pad C16.1 on F.Cu has no track and sits 0.150 mm outside the filled copper (zone clearance / thermal relief); connected to the pour. The relief spokes themselves are not modelled, so this pad reads slightly optimistic
- pad C44.1 on F.Cu has no track and sits 0.120 mm outside the filled copper (zone clearance / thermal relief); connected to the pour. The relief spokes themselves are not modelled, so this pad reads slightly optimistic
- via at (102.8, 98.1) lands on 4 layers (F.Cu, In1.Cu, In2.Cu, B.Cu) - barrel split into series sub-spans
- via at (103.5, 98.1) lands on 4 layers (F.Cu, In1.Cu, In2.Cu, B.Cu) - barrel split into series sub-spans
- via at (102.1, 98.1) lands on 4 layers (F.Cu, In1.Cu, In2.Cu, B.Cu) - barrel split into series sub-spans
- via array of 13 at centroid (117.3462, 94.8077) spanning 5.630 mm modelled as a 20-rung ladder over a 13.652 mm wide pour (1.9:1 aspect)
- via array of 7 at centroid (117.3, 99.1) spanning 4.200 mm modelled as a 20-rung ladder over a 13.652 mm wide pour (1.9:1 aspect)
- U9.D appears on 3 pads (also 5, S); treated as one terminal, computed once
- U11.D appears on 3 pads (also 5, S); treated as one terminal, computed once

J11.B -> U9.D : 3.3 V in at 1 A gives 3.2946 V out (5.391 mV drop, 5.391 mV). Zone model ladder, 200 nodes, solved in 34 ms. Least-resistance path 6.891 mohm, whole network 5.287 mohm (23.3% lower - parallel copper carries current)

#	Element	Detail	R @20 C	% of path
1	trace	1.1899 mm x 0.5 mm on B.Cu	774.1 uohm	11.2%
2	via	In2.Cu -> B.Cu, hole 0.3 mm	155.7 uohm	2.3%
3	via	In1.Cu -> In2.Cu, hole 0.3 mm	1.378 mohm	20.0%
4	via	F.Cu -> In1.Cu, hole 0.3 mm	155.7 uohm	2.3%
5	trace	6.5284 mm x 0.5 mm on F.Cu	4.247 mohm	61.6%
6	pour	7.5397 mm x 13.652 mm on F.Cu (pour)	179.6 uohm	2.6%

branch at pour copper on F.Cu at -7.625 mm along the pour axis (and 16 more): the net is not a simple series chain - see the network figure for the effect of the parallel copper

J11.B -> C17.1 : 3.3 V in at 1 A gives 3.2946 V out (5.389 mV drop, 5.389 mV). Zone model ladder, 200 nodes, solved in 34 ms. Least-resistance path 6.888 mohm, whole network 5.285 mohm (23.3% lower - parallel copper carries current)

#	Element	Detail	R @20 C	% of path
1	trace	1.1899 mm x 0.5 mm on B.Cu	774.1 uohm	11.2%
2	via	In2.Cu -> B.Cu, hole 0.3 mm	155.7 uohm	2.3%
3	via	In1.Cu -> In2.Cu, hole 0.3 mm	1.378 mohm	20.0%
4	via	F.Cu -> In1.Cu, hole 0.3 mm	155.7 uohm	2.3%
5	trace	6.5284 mm x 0.5 mm on F.Cu	4.247 mohm	61.7%
6	pour	7.4350 mm x 13.652 mm on F.Cu (pour)	177.2 uohm	2.6%

branch at pour copper on F.Cu at -8.670 mm along the pour axis (and 15 more): the net is not a simple series chain - see the network figure for the effect of the parallel copper

J11.B -> C16.1 : 3.3 V in at 1 A gives 3.2946 V out (5.359 mV drop, 5.359 mV). Zone model ladder, 200 nodes, solved in 32 ms. Least-resistance path 6.841 mohm, whole network 5.255 mohm (23.2% lower - parallel copper carries current)

#	Element	Detail	R @20 C	% of path
1	trace	1.1899 mm x 0.5 mm on B.Cu	774.1 uohm	11.3%
2	via	In2.Cu -> B.Cu, hole 0.3 mm	155.7 uohm	2.3%
3	via	In1.Cu -> In2.Cu, hole 0.3 mm	1.378 mohm	20.1%
4	via	F.Cu -> In1.Cu, hole 0.3 mm	155.7 uohm	2.3%
5	trace	6.5284 mm x 0.5 mm on F.Cu	4.247 mohm	62.1%
6	pour	5.4708 mm x 13.652 mm on F.Cu (pour)	130.4 uohm	1.9%

branch at pour copper on F.Cu at -15.061 mm along the pour axis (and 10 more): the net is not a simple series chain - see the network figure for the effect of the parallel copper

J11.B -> R8.1 : 3.3 V in at 1 A gives 3.2935 V out (6.516 mV drop, 6.516 mV). Zone model ladder, 200 nodes, solved in 32 ms. Least-resistance path 8.019 mohm, whole network 6.391 mohm (20.3% lower - parallel copper carries current)

#	Element	Detail	R @20 C	% of path
1	trace	1.1899 mm x 0.5 mm on B.Cu	774.1 uohm	9.7%
2	via	In2.Cu -> B.Cu, hole 0.3 mm	155.7 uohm	1.9%
3	via	In1.Cu -> In2.Cu, hole 0.3 mm	1.378 mohm	17.2%
4	via	F.Cu -> In1.Cu, hole 0.3 mm	155.7 uohm	1.9%
5	trace	6.5284 mm x 0.5 mm on F.Cu	4.247 mohm	52.0%
6	pour	25.4531 mm x 13.652 mm on F.Cu (pour)	606.5 uohm	7.6%
7	trace	0.5480 mm x 0.254 mm on F.Cu	701.8 uohm	8.8%

branch at pour copper on F.Cu at -5.438 mm along the pour axis (and 26 more): the net is not a simple series chain - see the network figure for the effect of the parallel copper

J11.B -> C44.1 : 3.3 V in at 1 A gives 3.2948 V out (5.230 mV drop, 5.230 mV). Zone model ladder, 200 nodes, solved in 34 ms. Least-resistance path 6.713 mohm, whole network 5.129 mohm (23.6% lower - parallel copper carries current)

#	Element	Detail	R @20 C	% of path
1	trace	1.1899 mm x 0.5 mm on B.Cu	774.1 uohm	11.5%
2	via	In2.Cu -> B.Cu, hole 0.3 mm	155.7 uohm	2.3%
3	via	In1.Cu -> In2.Cu, hole 0.3 mm	1.378 mohm	20.5%
4	via	F.Cu -> In1.Cu, hole 0.3 mm	155.7 uohm	2.3%
5	trace	6.5284 mm x 0.5 mm on F.Cu	4.247 mohm	63.3%
6	pour	0.0853 mm x 13.652 mm on F.Cu (pour)	2.032 uohm	0.0%

branch at pour copper on F.Cu at -15.061 mm along the pour axis (and 5 more): the net is not a simple series chain - see the network figure for the effect of the parallel copper

J11.B -> CP2.1 : 3.3 V in at 1 A gives 3.2947 V out (5.286 mV drop, 5.286 mV). Zone model ladder, 200 nodes, solved in 33 ms. Least-resistance path 6.768 mohm, whole network 5.184 mohm (23.4% lower - parallel copper carries current)

#	Element	Detail	R @20 C	% of path
1	trace	1.1899 mm x 0.5 mm on B.Cu	774.1 uohm	11.4%
2	via	In2.Cu -> B.Cu, hole 0.3 mm	155.7 uohm	2.3%
3	via	In1.Cu -> In2.Cu, hole 0.3 mm	1.378 mohm	20.4%
4	via	F.Cu -> In1.Cu, hole 0.3 mm	155.7 uohm	2.3%
5	trace	6.5284 mm x 0.5 mm on F.Cu	4.247 mohm	62.8%
6	pour	2.3996 mm x 13.652 mm on F.Cu (pour)	57.17 uohm	0.8%

branch at pour copper on F.Cu at -15.061 mm along the pour axis (and 5 more): the net is not a simple series chain - see the network figure for the effect of the parallel copper

J11.B -> U2.CSP2 : 3.3 V in at 0.001 A gives 3.2998 V out (0.199 mV drop, 0.000 mV). Zone model mesh, 4193 nodes, solved in 89 ms. Least-resistance path 229.7 mohm, whole network 195.5 mohm (14.9% lower - parallel copper carries current)

#	Element	Detail	R @20 C	% of path
1	trace	1.1899 mm x 0.5 mm on B.Cu	774.1 uohm	0.3%
2	via	In2.Cu -> B.Cu, hole 0.3 mm	155.7 uohm	0.1%
3	via	In1.Cu -> In2.Cu, hole 0.3 mm	1.378 mohm	0.6%
4	via	F.Cu -> In1.Cu, hole 0.3 mm	155.7 uohm	0.1%
5	trace	6.5284 mm x 0.5 mm on F.Cu	4.247 mohm	1.8%
6	mesh	meshed pour, 108 cells	34.88 mohm	15.2%
7	trace	0.5480 mm x 0.254 mm on F.Cu	701.8 uohm	0.3%
8	trace	5.0870 mm x 0.1524 mm on F.Cu	10.86 mohm	4.7%
9	via	F.Cu -> In2.Cu, hole 0.3 mm	1.534 mohm	0.7%
10	trace	6.7152 mm x 0.1524 mm on In2.Cu	21.7 mohm	9.5%
11	via	In1.Cu -> In2.Cu, hole 0.3 mm	1.378 mohm	0.6%
12	trace	43.7937 mm x 0.1524 mm on In1.Cu	141.5 mohm	61.6%
13	via	In1.Cu -> B.Cu, hole 0.3 mm	1.534 mohm	0.7%
14	trace	5.5081 mm x 0.2032 mm on B.Cu	8.817 mohm	3.8%

branch at (F.Cu, 12, 29) (and 113 more): the net is not a simple series chain - see the network figure for the effect of the parallel copper

J11.B -> U2.BAT : 3.3 V in at 0.001 A gives 3.2999 V out (0.099 mV drop, 0.000 mV). Zone model mesh, 4193 nodes, solved in 92 ms. Least-resistance path 130 mohm, whole network 96.9 mohm (25.5% lower - parallel copper carries current)

#	Element	Detail	R @20 C	% of path
1	trace	1.1899 mm x 0.5 mm on B.Cu	774.1 uohm	0.6%
2	via	In2.Cu -> B.Cu, hole 0.3 mm	155.7 uohm	0.1%
3	via	In1.Cu -> In2.Cu, hole 0.3 mm	1.378 mohm	1.1%
4	via	F.Cu -> In1.Cu, hole 0.3 mm	155.7 uohm	0.1%
5	trace	6.5284 mm x 0.5 mm on F.Cu	4.247 mohm	3.3%
6	mesh	meshed pour, 105 cells	33.9 mohm	26.1%
7	trace	5.2213 mm x 0.254 mm on F.Cu	6.687 mohm	5.1%
8	via	F.Cu -> In1.Cu, hole 0.3 mm	155.7 uohm	0.1%
9	trace	38.9583 mm x 0.254 mm on In1.Cu	75.55 mohm	58.1%
10	via	In1.Cu -> B.Cu, hole 0.3 mm	1.534 mohm	1.2%
11	trace	3.4278 mm x 0.2032 mm on B.Cu	5.487 mohm	4.2%

branch at (F.Cu, 12, 29) (and 110 more): the net is not a simple series chain - see the network figure for the effect of the parallel copper

J11.B -> U11.D : 3.3 V in at 1 A gives 3.2945 V out (5.465 mV drop, 5.465 mV). Zone model ladder, 200 nodes, solved in 35 ms. Least-resistance path 8.581 mohm, whole network 5.36 mohm (37.5% lower - parallel copper carries current)

#	Element	Detail	R @20 C	% of path
1	trace	1.1899 mm x 0.5 mm on B.Cu	774.1 uohm	9.0%
2	via	In2.Cu -> B.Cu, hole 0.3 mm	155.7 uohm	1.8%

3	via	In1.Cu -> In2.Cu, hole 0.3 mm	1.378 mohm	16.1%
4	via	F.Cu -> In1.Cu, hole 0.3 mm	155.7 uohm	1.8%
5	trace	6.5284 mm x 0.5 mm on F.Cu	4.247 mohm	49.5%
6	pour	6.7490 mm x 13.652 mm on F.Cu (pour)	160.8 uohm	1.9%
7	via	F.Cu -> In1.Cu, hole 0.3 mm (in pour)	155.7 uohm	1.8%
8	via	In1.Cu -> In2.Cu, hole 0.3 mm (in pour)	1.378 mohm	16.1%
9	via	In2.Cu -> B.Cu, hole 0.3 mm (in pour)	155.7 uohm	1.8%
10	pour	0.8142 mm x 13.652 mm on B.Cu (pour)	19.4 uohm	0.2%

branch at pour copper on F.Cu at -8.670 mm along the pour axis (and 10 more): the net is not a simple series chain - see the network figure for the effect of the parallel copper

P1.B -> C15.1 : 3.3 V in at 0.05 A gives 3.2903 V out (9.661 mV drop, 0.483 mW). Zone model ladder, 200 nodes, solved in 33 ms. Least-resistance path 191.1 mohm, whole network 189.5 mohm (0.9% lower - parallel copper carries current)

#	Element	Detail	R @20 C	% of path
1	trace	1.1899 mm x 0.5 mm on B.Cu	774.1 uohm	0.4%
2	via	In2.Cu -> B.Cu, hole 0.3 mm	155.7 uohm	0.1%
3	via	In1.Cu -> In2.Cu, hole 0.3 mm	1.378 mohm	0.7%
4	via	F.Cu -> In1.Cu, hole 0.3 mm	155.7 uohm	0.1%
5	trace	6.5284 mm x 0.5 mm on F.Cu	4.247 mohm	2.2%
6	pour	25.4531 mm x 13.652 mm on F.Cu (pour)	606.5 uohm	0.3%
7	trace	0.5480 mm x 0.254 mm on F.Cu	701.8 uohm	0.4%
8	trace	5.0870 mm x 0.1524 mm on F.Cu	10.86 mohm	5.7%
9	via	F.Cu -> In2.Cu, hole 0.3 mm	1.534 mohm	0.8%
10	trace	6.7153 mm x 0.1524 mm on In2.Cu	21.7 mohm	11.4%
11	via	In1.Cu -> In2.Cu, hole 0.3 mm	1.378 mohm	0.7%
12	trace	43.7937 mm x 0.1524 mm on In1.Cu	141.5 mohm	74.1%
13	via	In1.Cu -> B.Cu, hole 0.3 mm	1.534 mohm	0.8%
14	trace	2.8427 mm x 0.2032 mm on B.Cu	4.551 mohm	2.4%

branch at pour copper on F.Cu at -5.438 mm along the pour axis (and 10 more): the net is not a simple series chain - see the network figure for the effect of the parallel copper

P1.B -> C39.2 : 3.3 V in at 0.05 A gives 3.2954 V out (4.618 mV drop, 0.231 mW). Zone model ladder, 200 nodes, solved in 34 ms. Least-resistance path 92.2 mohm, whole network 90.57 mohm (1.8% lower - parallel copper carries current)

#	Element	Detail	R @20 C	% of path
1	trace	1.1899 mm x 0.5 mm on B.Cu	774.1 uohm	0.8%
2	via	In2.Cu -> B.Cu, hole 0.3 mm	155.7 uohm	0.2%
3	via	In1.Cu -> In2.Cu, hole 0.3 mm	1.378 mohm	1.5%
4	via	F.Cu -> In1.Cu, hole 0.3 mm	155.7 uohm	0.2%
5	trace	6.5284 mm x 0.5 mm on F.Cu	4.247 mohm	4.6%
6	pour	25.6627 mm x 13.652 mm on F.Cu (pour)	611.5 uohm	0.7%
7	trace	5.2213 mm x 0.254 mm on F.Cu	6.687 mohm	7.3%
8	via	F.Cu -> In1.Cu, hole 0.3 mm	155.7 uohm	0.2%
9	trace	38.9983 mm x 0.254 mm on In1.Cu	75.55 mohm	81.9%
10	via	In1.Cu -> B.Cu, hole 0.3 mm	1.534 mohm	1.7%
11	trace	0.7325 mm x 0.25 mm on B.Cu	953.1 uohm	1.0%

branch at pour copper on F.Cu at -5.438 mm along the pour axis (and 27 more): the net is not a simple series chain - see the network figure for the effect of the parallel copper